

Microsoft Exchange From Deserialization to Post-Auth RCE (CVE-2021-28482)

Microsoft Exchange From Deserialization to Post-Auth RCE (CVE-2021-28482) - Jang, Medium.

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After the proxylogon event that happened last March, it seems to have given a new source of inspiration to Researchers who have worked or are working on Exchange.

This attack vector is quite new, it does not follow the well-worn paths of Exchange vulnerabilities; probably nobody had found such an attack chain on Exchange before.

Also at that time Pwn2Own 2021 Vancouver was taking place, and in the target list Exchange carried a prize of 200k\$USD ~ = 5 billion VND ...

This was probably the hottest target in this p2o round, with up to 3 teams targeting Exchange, among them: Orange from Devcore, Pham Khanh from Viettel, Steven from SrcIncite.

Even though on the very first attempt team Orange succeeded and took the whole 200k\$, the heat on the following days whenever the Exchange target came up did not drop at all. Me, my colleagues, and like so many other researchers were just waiting for the Exchange target to go live, to satisfy our curiosity and to see whether the author would drop any small piece of information during the PoC ☐. Although the 2 later bugs were both marked as Partial, probably because they duplicated orange's first bug, we still greatly admire them; the bad luck was only a matter of who went first, since the caliber of the 3 chains the 3 teams brought to attack was all the same.

One week later, on 13/04, M\$ released the patch. It was a bit of a surprise that in this release up to 2 of the 4 Exchange CVEs were pre-auth, and all 4 of those CVEs could lead to RCE. The surprise is that p2o had just taken place the week before, and this week there was already a patch, so this may be a completely different bug from the bunch of bugs used at that p2o.

#DIFF PATCH

After the proxylogon affair, Exchange suddenly seems to have become hot again.

When the patch came out, many parties rushed in to tear the patch apart, from the black-hat world to the red-hat world; everywhere put up to 80% of their manpower into this.

And of course, I was no exception (◕‿◕).

Initially my goal with my colleague was to research the 2 pre-auth bugs first; probably many other parties did the same.

However this was not as simple as we thought, and we did not succeed in analyzing these 2 pre-auth bugs,

Below are just some notes about the 2 pre-auth bugs CVE-2021-28480 and 28481:

In *BackEndCookieEntryParser.TryParse()*, a few new lines of code were added to re-check the host name and FQDN:

It is possible that a bug like the X-BEResource-Cookie of proxylogon also existed here, which could be abused to set BackEndServer and then SSRF into the backend.

This class handles cookies of the form: *X-BackEndCookie/X-BackEndCookie2*.

BackEndCookieEntryParser.TryParse() -> *UnObscurify()*. This method simply works by base64 decoding the cookie and xoring it with the char "0xff":

After decoding, a string from X-BackEndCookie has the following form:

```
**Database~**eb60615b-fc77-44b7-b0e4-a7abf6f7f57e**~**2021-05-20T01:48:22
```

Besides the Database kind, there is another kind, **Server**:

Of the following form:

```
**Server**~**exchange.evill.corp**~1942062522~2021-05-19T08:36:11
```

It looks very much like the previous X-BEResource bug,

However, to abuse this X-BackEndCookie bug, not every entrypoint will do!

This cookie is only received and handled by classes that inherit the class **BEServerCookieProxyRequestHandler**:

Some of its inheriting classes are:

That is not the end of it, the story still continues!

The classes above all inherit ProxyRequestHandler, and after each successful computation of the BackEnd value, the method *DoProtocolSpecificRoutingTargetOverride()* -> *RedirectIfNeeded()* is called:

Among them, the class *OwaEcpProxyRequestHandler* overrides this *RedirectIfNeeded()* method:

It handles and looks for the existence of the FQDN in the system, and if it is not there it throws an exception right away.

This is the reason why many parties also discovered and tried this BackendCookie but failed at the endpoints "/ecp, /owa".

To handle it we need to find classes that inherit the class

BEServerCookieProxyRequestHandler, but whose `*RedirectIfNeeded()` method must also satisfy the requirement of not touching that FQDN value.

Among that bunch I found a few feasible classes as follows:

```
- AnonymousCalendarProxyRequestHandler
- ComplianceServiceProxyRequestHandler
- EwsAutodiscoverProxyRequestHandler
- MailboxDeliveryProxyRequestHandler
- MapiProxyRequestHandler
- MicroServiceProxyRequestHandler
- MrsProxyRequestHandler
- OabProxyRequestHandler
...
```

However, for the unauthenticated case, I have only been able to reach

AnonymousCalendarProxyRequestHandler and **EwsAutodiscoverProxyRequestHandler**.

Although the FQDN can now be set arbitrarily, before making the request ProxyRequestHandler checks `Host == BackendServer.Fqdn` once again.

If you deliberately change the FQDN to an invalid value to obtain SSRF as in ProxyLogon, an exception is thrown at this point!

Therefore, all we can do now is change it to some valid FQDN.

For **AnonymousCalendarProxyRequestHandler**:

The host can be arbitrary, but an error occurs because the server cannot generate the packet needed to authenticate to the backend. This means it can only be changed to a backend belonging to the system.

For **EwsAutodiscoverProxyRequestHandler**:

A request can be sent to an arbitrary host. However, it contains the Anonymous user's token, which means it has very few privileges if sent to the backend.

My investigation of the backend cookie bug stopped here for more than a week, and I still have not made any further progress.

That is what I wanted to share about these two pre-authentication bugs.

Next is the post-authentication RCE bug, CVE-2021-28482:

This patch removed two files from the Exchange server:

- *Microsoft.Exchange.Clients.Owa2.Server.Web.MeetingPollHandler*
- *Microsoft.Exchange.HttpProxy.PsgwProxyRequestHandler*

I searched everywhere for Psgw but could not find an entry point that would let me reach it, so I turned to *MeetingPollHandler*.

From the web.config for ClientAccess/owa, we can identify the way to reach this entry point: */owa/MeetingPollHandler.ashx*:

The code that handles MeetingPollHandler is as follows:

```
MeetingPollHandler.ProcessRequest()-> MeetingPollProposeOptionsPayload.ProcessRequest()
```

The more interesting part is inside the **MeetingPollProposeOptionsPayload.GetRequests()** method.

Here, this method calls *EntitySerializer.Deserialize()*, and *EntitySerializer.Deserialize()* in turn calls *DataContractSerializer.ReadObject()*.

I have worked extensively on Java deserialization, but had never worked on .NET deserialization before.

However, through research and discussions with others, I learned that RCE with *DataContractSerializer* is entirely possible if you can control the type used for deserialization, or if that type has some weakness that can be exploited.

This is the second case: the type being deserialized is too permissive. Let us look more closely.

The data is deserialized as follows:

```
EntitySerializer.**Deserialize**<Dictionary<string,  
**ProposeOptionsMeetingPollParameters**>>(largeStringProperty);
```

The **ProposeOptionsMeetingPollParameters** class inherits from **SchematizedObject**. **SchematizedObject** inherits from **PropertyChangeTrackingObject**.

The key to this vulnerability lies in **PropertyChangeTrackingObject** itself:

This class also has a nested class containing information about entities. Those entities are stored in the **ChangedProperties** field, whose type is **PropertyBag**.

PropertyBag has just one DataMember field, of type **Dictionary<string, object>**:

~\(\)/~

Following this logic, when MeetingPollHandler deserializes the type *Dictionary<string, ProposeOptionsMeetingPollParameters>*, I can insert a gadget chain into the *propertyValues* field within *ChangedProperties* during deserialization. This is quite similar to Java deserialization!

The logic is illustrated in the following diagram:

The gadget chain I used is *ObjectDataProvider*, generated by ysoserial.net.

However, several adjustments are needed to match how Exchange Server operates:

The next step is to create a meeting and insert the gadget into it. Documentation on creating the meeting can be found by searching.

Finally, trigger it with a URL of the following form:

Here, XXXX is the ID of the meeting containing the payload.

After triggering it through MeetingPollHandler, cmd is spawned on the Exchange server with the OWA application's w3wp as its parent process:

PoC:

This is only one of several entry points that can trigger this vulnerability. Other entry points may trigger it without this interaction. However, I have not investigated those thoroughly or produced a PoC for them at this time.

I hope this article helps resolve some of the difficulties other researchers are encountering. Perhaps they will find a better direction to pursue ;).

Thanks for reading,

Jang of VNPT ISC

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